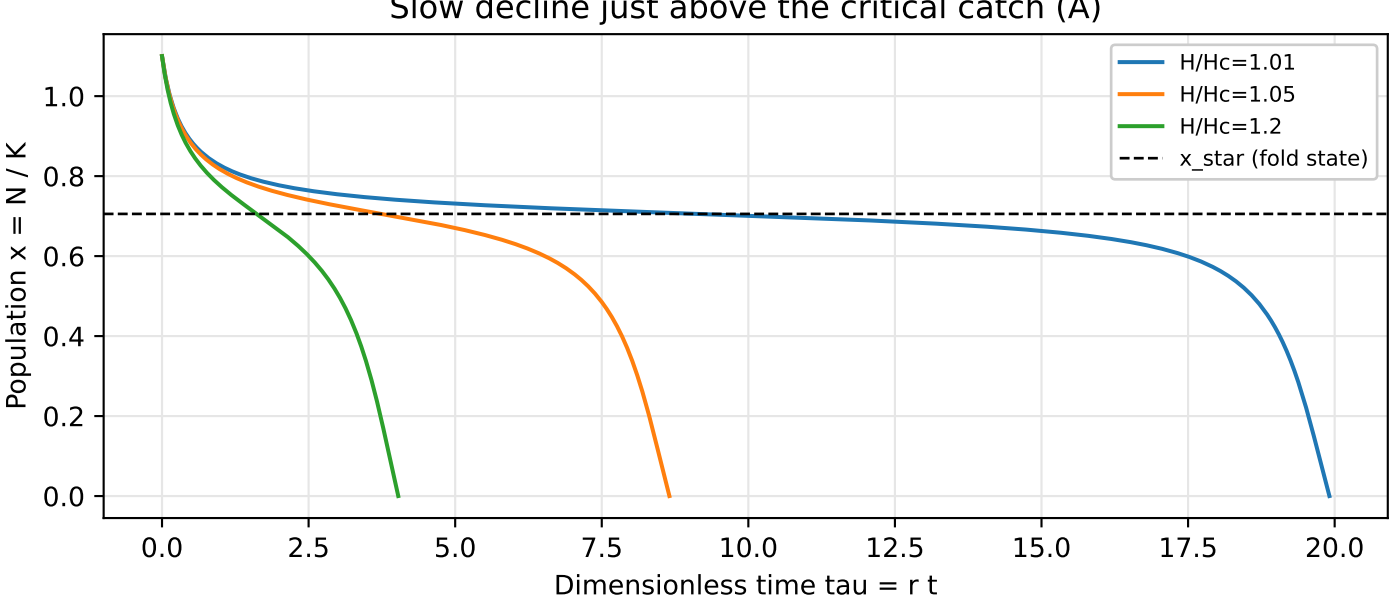Synthetic model: r=0.7, K=100, A=20; declared model units